Size Estimation

Estimate based upon what info?

Statistics

- Size – based upon structure
- Value – based upon data

Use statistics to estimate condition selectivity

Size statistics per table

	#tuples
	#blocks
	#bytes per tuple
	#tuples that fit into a block (*blocking factor*)

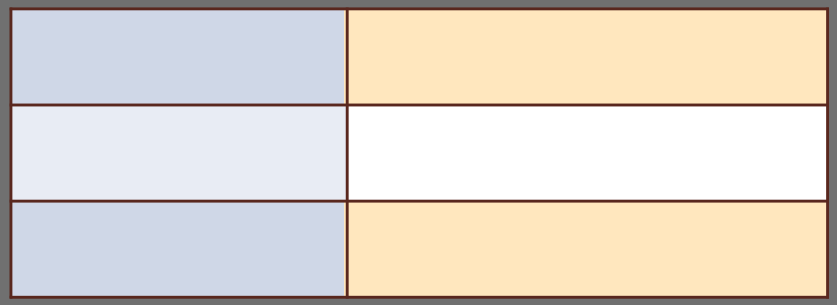
Ignoring variable-length types, like TEXT.

If no fragmentation, $B(R) = \left\lceil \frac{T(R)}{bf(R)} \right\rceil$.

Size estimates

	Cross		
	Inner		If joining on FK from R to S.
			Otherwise
	Outer		If joining on FK from R to S.
			Otherwise
	Left		If joining on FK from R to S.
			Otherwise
	Anti		

Size estimates



Block size of joined tables

		If joining on FK from R to S.

Recall:

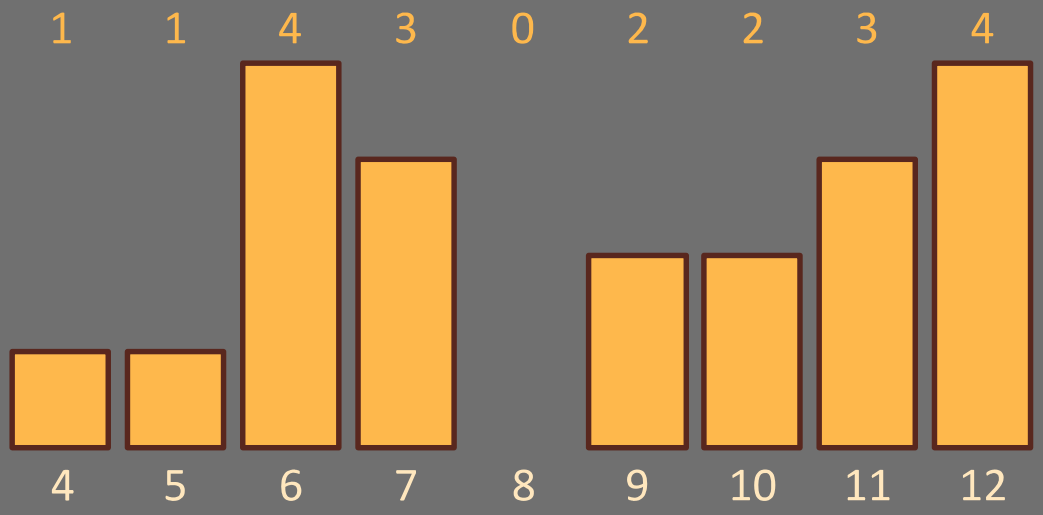
- $len(X) \cdot T(X) = B(X) \cdot bs$
- $len(XY) = len(X) + len(Y)$

$B(XY) = ?$

Value statistics for attribute R.a

	#distinct values
	minimum value
	maximum value
histogram	Summarize value distribution

Histogram



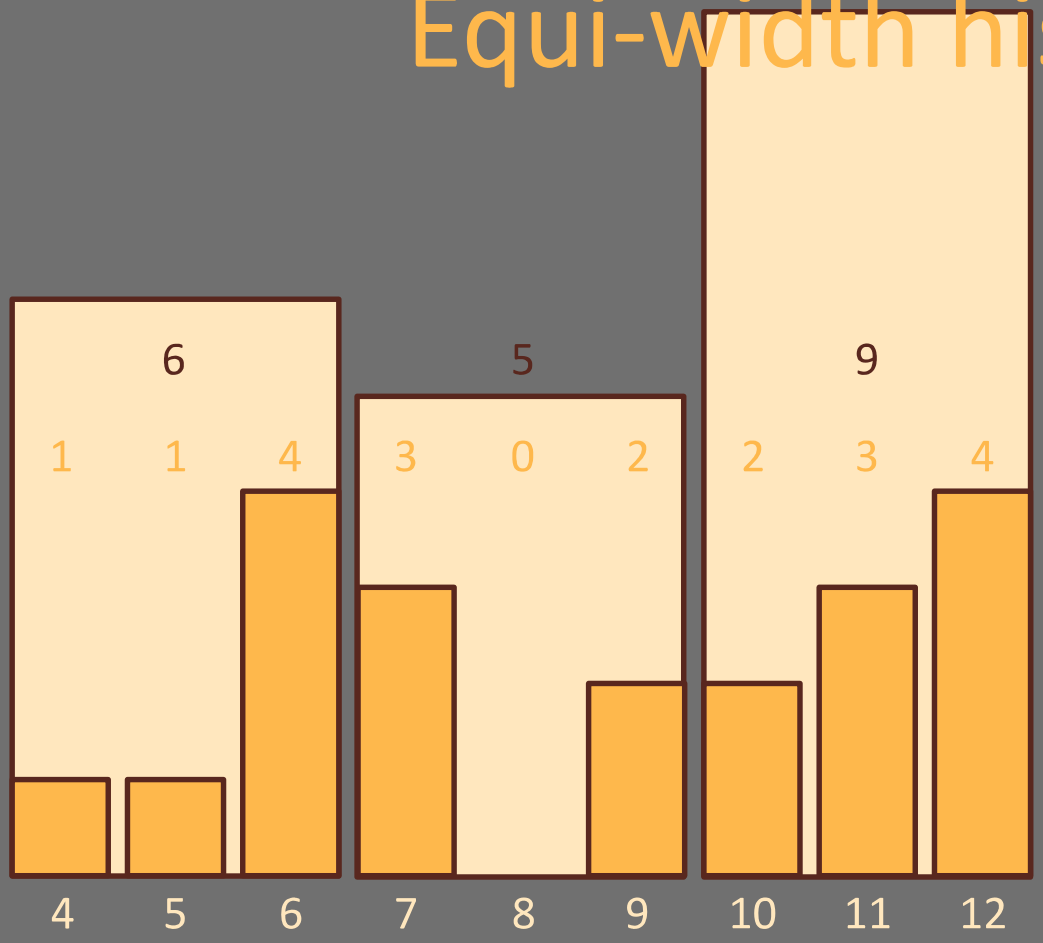
Storage space:

#values × (size(value) + size(int))

OR if values are sequential ints,

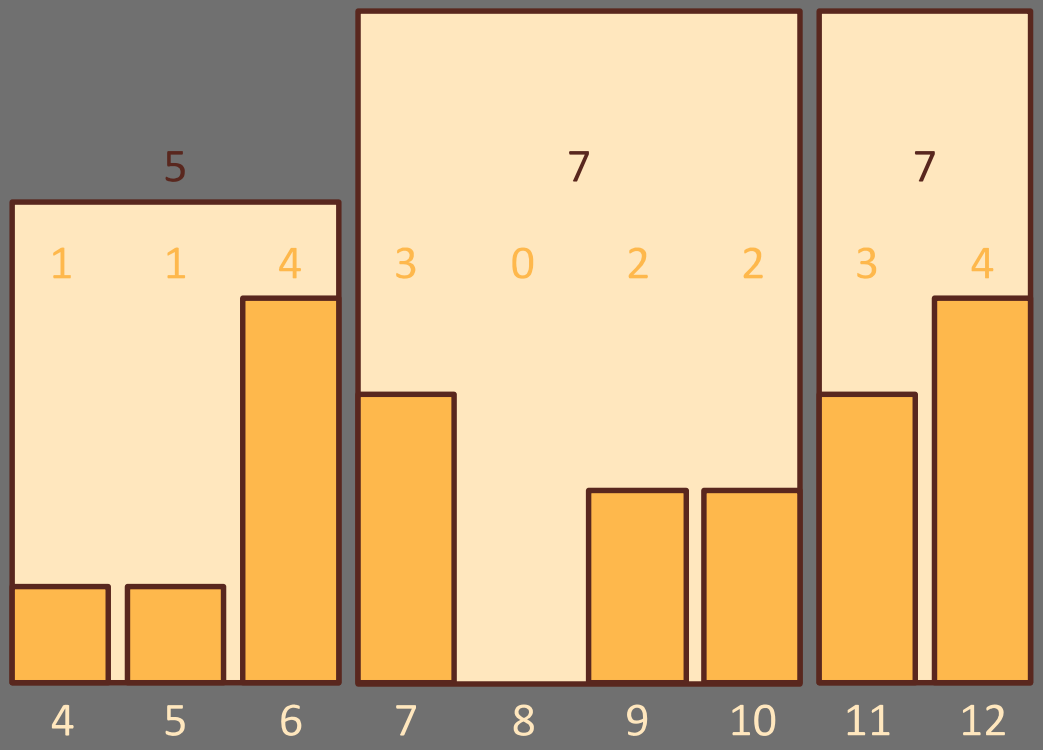
(2 + #values) × size(int)

Equi-width histogram



Storage space:
if values are sequential ints,
(2 + #boxes) × size(int)

Equi-depth histogram



Storage space:
if values are sequential ints,
(2 × #boxes + 1) × size(int)

Can be more useful when
distribution is non-uniform.

Histograms summary

- Several types
- Goals
  - Efficient to compute
  - Useful in estimating condition selectivity

Managing statistics

- Accuracy vs. resources
  - Update frequency
  - Histograms for which attributes? How accurate?
- Computed automatically
- Some manual control: `ANALYZE`
- Can view statistics: `SELECT * FROM pg_stats`